Monologue or Dialogue?

Measuring Conversational Structure in the Reasoning Traces of Large Language Models

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Abstract

A reasoning model such as DeepSeek-R1 produces a long "thinking" trace before it gives an answer. That trace can be read as one mind working step by step, or as several voices exchanging proposals and objections. This paper asks which reading the text itself supports once it is measured. Reasoning models are recent, so there is little prior work on the voice structure of their traces, and no trace comes labelled with its true structure; we therefore compare against a small grid of corpora whose structure is known: human and AI, monologue and dialogue. We test three text-analysis methods and carry forward those that correctly classify all four known corpora as monologue or dialogue. Two of the three do so; the third, a coarse document-level author count, fails to recognise multiple voices even in a genuine four-way debate, so it does not judge the traces. Both carried-forward methods place the traces on the dialogue side of the scale, most clearly in how the writing style shifts from sentence to sentence and in how often a question is followed by an answer. We are explicit about how much weight each method can bear, and flag that one of the two depends on an automatic labeller of only moderate reliability.

1 Introduction

Reasoning-oriented training has become standard in frontier models. Given a problem, a model such as DeepSeek-R1 [1] first produces a long stretch of intermediate text, often called a reasoning trace [2], and only then gives its final answer. These traces have a notable feature: each is written by a single model in one pass, yet a trace often contains questions and their answers, objections and concessions, and sudden reversals such as "Wait, that is not right." In writing by people, features like these can mark a real exchange between participants, though they can also mark a single inner voice. This raises the question: is a reasoning trace's textual shape closer to that of a monologue or of a dialogue?

Kim et al. [3] argue that reasoning-trained models improve in part because they simulate a "society of thought," a set of internal perspectives that propose, criticise, and reconcile. There are interesting parallels in older ideas from cognitive science, which have variously understood reasoning as a social and argumentative activity [4] and the mind as a set of interacting agents [5]. Our study works from an assumption: if a trace is produced by several voices taking turns, that should leave marks in the writing. Authorship analysis rests on the observation that writers differ in small, largely involuntary habits, most robustly in how they use function words, and that these habits are stable within an author and vary between authors [6]. We therefore treat the question in operational terms: a monologue should read as the work of one author and a dialogue as the work of several authors, and we ask which of the two the traces are closer to.

2 Study design

Reasoning models are new enough that there is little prior work on the voice structure of their traces, and no collection of traces comes with its structure labelled. Our approach is to compare against a grid of corpora whose structure *is* known (Figure 1). We assemble four such corpora and vary two properties across them: who produced the text (a human or an AI) and whether it is a monologue or a dialogue. The four cells are human monologue (think-aloud protocols, in which a person speaks their thoughts aloud while solving a task [7]), human dialogue (group deliberation from DeliData [8]), AI monologue (a single Qwen-2.5-7B model solving a puzzle on its own), and AI dialogue (four separate copies of Qwen-2.5-7B, each given a role, taking turns in a debate). The fifth cell is DeepSeek-R1 itself: the model is given the same puzzle, and we take its native thinking trace — the reasoning it produces before its final answer — as the document. This is the cell whose structure we want to determine.

The grid functions as a validity check. We define the check at the level of the cell and apply it identically to every method: does the method place each of the four known cells on the correct side of the monologue–dialogue axis? A method is carried forward to the traces only if it classifies all four known cells correctly. The rationale runs both ways: a method that cannot recognise multiple voices in a genuine four-way debate is in no position to rule them out in a reasoning trace, and a method that reports several voices in one person's think-aloud protocol is in no position to affirm them in a trace either.

We hold the task fixed where possible. The AI monologue, AI dialogue, and DeepSeek-R1 cells all solve the same puzzle, Game of 24 [9], a small arithmetic puzzle in which four given numbers must be combined with the four basic operations to reach 24, and the human monologue cell uses the same puzzle. This rules out a method separating the cells by their subject matter rather than by their voice structure. The human dialogue cell is the one exception, since it uses the Wason card task, another compact logic puzzle,

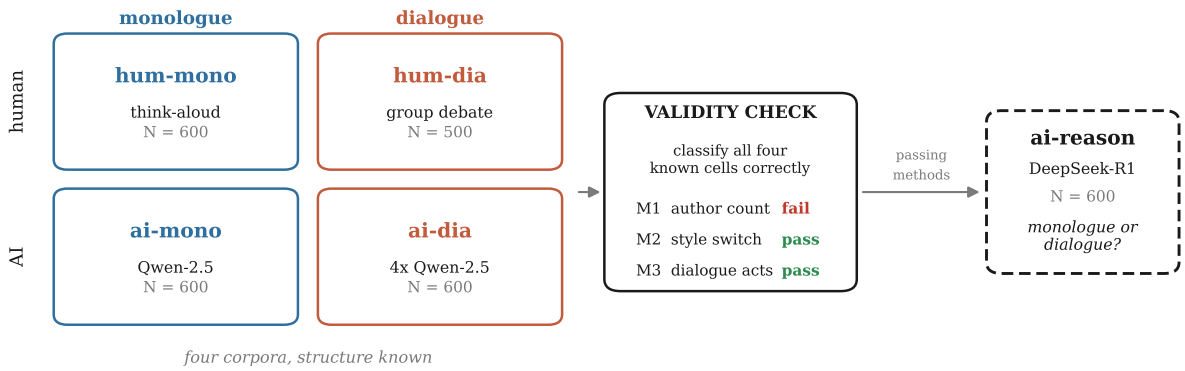


Figure 1: Study design. Four corpora of known voice structure (left) provide a validity check: we carry forward the methods that correctly classify all four known cells as monologue or dialogue. Two of the three methods do so; the coarse document-level author count does not, and is set aside. Carried-forward methods are applied to the DeepSeek-R1 traces (`ai-reason`) and place them on a monologue–dialogue scale anchored by the two same-task AI cells; the human cells serve only to check that a method works.

because no group-deliberation corpus exists for Game of 24; its role is only to test whether a method can detect human dialogue at all, so it never anchors the placement. We also removed speaker names, role titles, and letter labels from both dialogue corpora before analysis, so that a method cannot detect dialogue by identifying who is speaking instead of reading the structure of the exchange.

3 Methods

The three methods read deliberately different signals, so that agreement among them carries more weight than any single method alone. Each turns its signal into a monologue-or-dialogue call for a cell; Section 4 reports which methods pass the validity check and what they say about the traces.

Method 1: counting authors across the document [10]. This method treats a document as a whole. It divides the text into paragraph-sized windows, describes each window by how often the 50 most common words occur in it, and groups similar windows together. The number of groups is read as the number of authors, which we reduce to a binary decision: one author (monologue) or more than one (dialogue). A cell counts as correctly classified if its documents fall on the right side of that decision on balance.

Method 2: detecting style changes between sentences [11]. This method operates sentence by sentence. For each pair of neighbouring sentences it estimates the probability that the two were written by different authors, using an encoder trained to capture writing style while ignoring topic [12]. Longer documents contain more sentence pairs, which affects the raw score, so we adjust for length before comparing cells. Each document receives a single number, which we call its *lean*, indicating whether its style-change pattern lies closer to the dialogue cells or to the monologue cells; a cell is classified by the direction of its average lean.

Method 3: measuring question-and-answer structure [13, 14]. This method reads the text as a sequence of conversational moves. A language model tags each utterance with its function using CAMS, a standard scheme whose labels include a question, an answer, an agreement, or a suggestion. From these tags we measure how often an *adjacency pair* is completed. An adjacency pair is a two-part exchange in which the first part makes the second one due, such as a question followed by an answer, or a proposal followed by its acceptance. A single writer has little reason to produce completed pairs, so their presence is evidence of genuine dialogue; a cell is classified by whether its dialogue documents complete more pairs than its monologue documents. Because Method 3's labels are produced by a language model acting as a tagger, their reliability is a separate concern, which we measure against a human annotator and return to in Sections 4 and 6.

4 Results

The validity check. Of the three methods, two — Method 2 and Method 3 — correctly classify all four known cells as monologue or dialogue, and are carried forward to the traces. Method 1 does not: it misclassifies both dialogue cells, and is set aside. We take each method in turn.

Method 1 fails the check. Read as a summary number, its balanced accuracy is 0.6088 (95% confidence interval 0.596 to 0.621), above the chance level of 0.50; but that single number hides a one-sided failure. On single-author text the method is nearly perfect, correctly labelling 0.975 of the human monologue documents and 0.998 of the AI monologue documents. On multi-author text it collapses: it recovers only 0.423 of the AI debates — so a majority of genuine four-way debates are read as a single author — and *none* of the human debates, calling all 500 of them single-author. In cell terms, it places both known dialogue cells on the monologue side. A method that cannot see four speakers in a four-speaker debate, and that never once flags the human debate as multi-voiced, has not shown it can detect voices at all, so it does not judge the traces.

A note on the deviation from pre-registration. We originally pre-registered a separate statistical gate for each method rather than the single cell-level check used here. Under those per-method gates, Method 1's balanced accuracy cleared a chance baseline and would have "passed," while Method 3 was blocked by a label-reliability threshold ($\kappa \geq 0.70$) discussed below. On inspection those gates behaved pathologically: they admitted a method that labels genuine four-way debate as single-author, and excluded a

method that separates all four known cells correctly. The per-method gates were only ever an operationalisation of the idea stated from the outset — that a method must separate monologue from dialogue on the known corpora before it is trusted on the traces — and that operationalisation did not match the idea. We therefore report against the simpler, uniform criterion we set out to use: does the method correctly classify all four known cells? This is a stated deviation, not a silent one; the pre-registered gates and the reason for departing from them are recorded here so a reader can weigh both.

Method 2 passes the check cleanly. All four cells fall on the correct side: the monologue cells lean negative and the dialogue cells positive (average lean -0.0105 for the monologue cells against 0.0125 for the dialogue cells), and the separation is strong, with Cliff's $\delta = 0.6211$ (Cliff's δ is a rank-based measure of how cleanly two groups separate, from 0 at complete overlap to 1 at no overlap). Applied to the traces, Method 2 places them further toward dialogue than either AI anchor.

Method 3 passes the check, with a caveat about its labels. On the classification the check asks for, it performs well. Its completion rate separates dialogue from monologue in the expected direction (Cliff's $\delta = 0.4717$): within each producer, the dialogue cell completes more pairs than the monologue cell — 0.235 against 0.096 for the human cells, and 0.025 against 0.003 for the AI cells. Applied to the traces, its completion rate of 0.035 lies closer to the AI dialogue anchor (0.025) than to the AI monologue anchor (0.003), a dialogue-side placement.

The caveat is that this classification rests on labels from an automatic tagger, and separating the cells is not the same as labelling every utterance correctly: a tagger could be systematically wrong in a consistent way and still separate the cells. We measured the tagger against a human annotator using Cohen's κ, an agreement score that corrects for chance matches, and obtained $\kappa = 0.54$ — below the 0.70 we had pre-registered. We do not treat this as disqualifying, for a reason we can state but not fully settle here. CAMS uses 28 fine-grained labels, many of them close relatives such as `inform` and `suggest` that are easily confused even by careful human coders, and unweighted κ charges a near miss between two similar tags as heavily as a disagreement between two unrelated ones. The CAMS authors make the same point and recommend a measure that gives partial credit for near misses; under such a measure agreement rises to a tree-weighted 0.62, and the raw share of labels on which tagger and human agree is 0.74. These figures argue that the labels are *reasonable*, not that they are validated: they remain below our threshold, and we do not claim the labels are error-free. The honest resolution — an adjudicated double-annotation — is left to future work (Section 6). We therefore carry Method 3 forward on the strength of its cell-level separation, while flagging that its label reliability is the weakest link in its chain.

Across the methods (Table 1 and Figure 2), the two methods that pass the validity check agree: both place the traces on the dialogue side. Method 2, reading style changes between sentences, places them firmly past the dialogue anchor; Method 3, reading completed question-and-answer pairs, places them past it too. The one method that pointed toward monologue, Method 1, is not a dissenting vote we weigh against these two — it is a method that failed the check, and its reading of the traces is reported only to show it was not quietly dropped. For what it is worth, even Method 1 places the traces in ambiguous territory rather than cleanly at the monologue pole, but its verdict carries no weight, because it cannot classify the known dialogue cells to begin with.

5 Discussion

The claim here concerns the surface of the text, that is, the patterns a reader or a tool can observe in the writing. It says nothing directly about what occurs inside the model. On that surface, the picture leans toward dialogue. The traces change writing style from sentence to sentence in the manner of dialogue, and they complete question-and-answer pairs at a rate above the dialogue anchor.

The scope of this claim is limited. A dialogue-leaning reading does not show that a model contains several minds, just as a novel full of dialogue does not have several authors. It shows that the trace is organised like a conversation, which is a measurable fact in itself, and a natural surface counterpart to the internal multi-agent behaviour that Kim et al. [3] describe from the inside, through interpretability. The outside view and the inside view point the same way, which is reassuring, though the present evidence stands on its own as a statement about surface structure.

The strength of the finding survives even a sceptical reading of Method 3. Taken together, the two validated methods agree that the traces are dialogue-like, and that is the case we make. But suppose a reader sets Method 3 aside entirely on the grounds that its labels are only moderately reliable. The validated evidence then reduces to Method 2 — the method with the cleanest validation — which still places the traces on the dialogue side, and *no* validated method places them on the monologue side. (Method 1 does, but it failed the validity check and is not part of the validated evidence.) So the dialogue lean does not hinge on trusting the tagger: it holds with Method 3 and survives without it.

Two caveats temper the dialogue placements, and both concern how a distance was measured rather than its direction. Method 2's distance to the monologue side is partly an extrapolation, because the traces are much longer than the monologue anchor, so it is more accurate to say that they track the dialogue anchor closely over the range where the two overlap. Method 3's verdict rests on its completion signal, since its coarser tag-frequency profiles cannot separate the two AI anchors. Neither caveat changes a direction; each limits only how far past the pole the placement should be read.

6 Limitations

Several limitations should be noted. The target is a single reasoning model, DeepSeek-R1, on a single family of puzzles, so the verdict concerns this setting and does not yet generalise to reasoning traces at large. The two anchors were generated with a different base model (Qwen-2.5-7B)

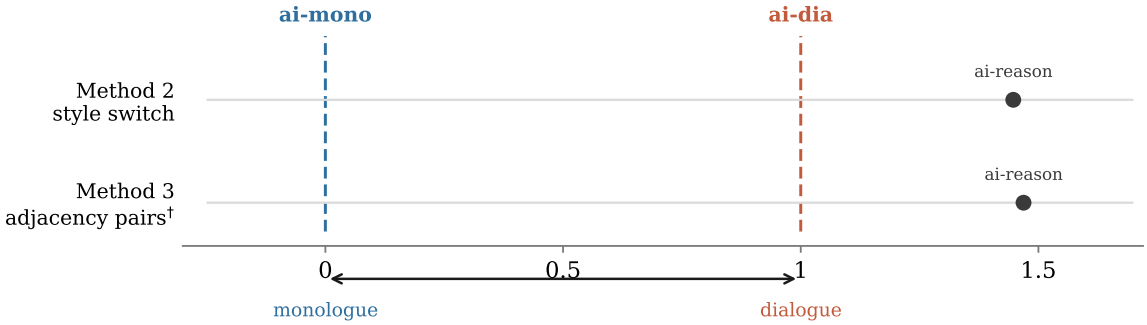


Figure 2: Placement of the traces by each carried-forward method on the monologue–dialogue scale, anchored so `ai-mono`= 0 and `ai-dia`= 1. Both methods place the DeepSeek-R1 traces past the dialogue pole and at nearly the same point (1.45 and 1.47); Method 3 is daggered to mark that its placement depends on an automatic labeller of only moderate reliability (Section 4). Method 1 is not shown here: it failed the validity check and does not place the traces (Table 1).

Table 1: Each method against the single validity check — does it place all four known cells on the correct side of the monologue/dialogue axis? — and, for the two that pass, where it places the traces. Method 3 passes the check but depends on an automatic labeller of moderate reliability (†; Section 4).

Method	Classifies all four known cells?	Carried forward?	Verdict on traces
1: author count	no	no	—
2: style switch	yes	yes	dialogue
3: adjacency pairs	yes	yes†	dialogue†

than the target, so part of the measured distance may reflect the change of model as well as voice structure, and the placement is best read as "closer to which pole" than as an exact distance. Method 2's monologue comparison rests on extrapolation, as noted above.

Method 3's labels come from an automatic tagger whose agreement with a human annotator ($\kappa = 0.54$ unweighted; tree-weighted 0.62; raw agreement 0.74) is only moderate. The validity check we apply confirms that the tagger *separates* the known cells in the right direction, but not that every label is correct, and these two are distinct; a systematic labelling bias could separate the cells without the labels being trustworthy in detail. The rigorous remedy is an adjudicated double-annotation, with two independent coders and a resolution step, which we leave to future work. Until then, Method 3 should be read as corroborating evidence rather than as an independently validated labelling.

Finally, a note on method more than on data. We report against a single cell-level validity check applied identically to all three methods, having departed from a more elaborate set of per-method gates fixed in advance (Section 4). The reason for the departure is stated rather than hidden, and the uniform check is the sharper instrument: a method earns the right to judge the traces exactly when it can tell monologue from dialogue on data whose structure we already know.

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